An update to "On the sign patterns of entrywise positivity preservers in fixed dimension"

Terence Tao · 30 September 2017

https://terrytao.wordpress.com/2017/09/30/an-update-to-on-the-sign-patterns-of-entrywise-positivity-preservers-in-fixed-dimension/

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Apoorva Khare and I have updated our paper "On the sign patterns of entrywise positivity preservers in fixed dimension", announced at this post from last month. The quantitative results are now sharpened using a new monotonicity property of ratios $s_\lambda(u)/s_\mu(u)$ of Schur polynomials, namely that such ratios are monotone non-decreasing in each coordinate of u if u is in the positive orthant, and the partition λ is larger than that of μ. (This monotonicity was also independently observed by Rachid Ait-Haddou, using the theory of blossoms.) In the revised version of the paper we give two proofs of this monotonicity. The first relies on a deep positivity result of Lam, Postnikov, and Pylyavskyy, which uses a representation-theoretic positivity result of Haiman to show that the polynomial combination

$$s_{(\lambda\wedge\nu)/(\mu\wedge\rho)} s_{(\lambda\vee\nu)/(\mu\vee\rho)} - s_{\lambda/\mu} s_{\nu/\rho} \tag{1}$$

of skew-Schur polynomials is Schur-positive for any partitions λ, μ, ν, ρ (using the convention that the skew-Schur polynomial $s_{\lambda/\mu}$ vanishes if μ is not contained in λ, and where $\lambda \wedge \nu$ and $\lambda \vee \nu$ denotes the pointwise min and max of λ and ν respectively). It is fairly easy to derive the monotonicity of $s_\lambda(u)/s_\mu(u)$ from this, by using the expansion

$$s_\lambda(u_1, \dots, u_n) = \sum_k u_1^k s_{\lambda/(k)}(u_2, \dots, u_n)$$

of Schur polynomials into skew-Schur polynomials (as was done in this previous post).

The second proof of monotonicity avoids representation theory by a more elementary argument establishing the weaker claim that the above expression (1) is non-negative on the positive orthant. In fact we prove a more general determinantal log-supermodularity claim which may be of independent interest:

> **Theorem 1** Let A be any $n \times n$ totally positive matrix (thus, every minor has a non-negative determinant). Then for any k-tuples I_1, I_2, J_1, J_2 of increasing elements of $\{1, \dots, n\}$, one has
>
> $$\det(A_{I_1\wedge I_2, J_1\wedge J_2}) \det(A_{I_1\vee I_2, J_1\vee J_2}) - \det(A_{I_1,J_1}) \det(A_{I_2,J_2}) \geq 0$$
>
> where $A_{I,J}$ denotes the $k \times k$ minor formed from the rows in I and columns in J.

For instance, if A is the matrix

$$A = \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix}$$

for some real numbers $a, \dots, p$, one has

$$ah - de \geq 0$$

(corresponding to the case $k = 1$, $I_1 = (1), I_2 = (2), J_1 = (4), J_2 = (1)$), or

$$\det \begin{pmatrix} a & c \\ i & k \end{pmatrix} \det \begin{pmatrix} f & h \\ n & p \end{pmatrix} - \det \begin{pmatrix} e & h \\ i & l \end{pmatrix} \det \begin{pmatrix} b & c \\ n & o \end{pmatrix} \geq 0$$

(corresponding to the case $k = 2$, $I_1 = (2,3), I_2 = (1,4), J_1 = (1,4), J_2 = (2,3)$). It turns out that this claim can be proven relatively easy by an induction argument, relying on the Dodgson and Karlin identities from this previous post; the difficulties are largely notational in nature. Combining this result with the Jacobi-Trudi identity for skew-Schur polynomials (discussed in this previous post) gives the non-negativity of (1); it can also be used to directly establish the monotonicity of ratios $s_\lambda(u)/s_\mu(u)$ by applying the theorem to a generalised Vandermonde matrix.

(Log-supermodularity also arises as the natural hypothesis for the FKG inequality, though I do not know of any interesting application of the FKG inequality in this current setting.)

Comments (9)



Fred Lunnon · 30 September 2017

Formulae preceding “corresponding to the case {k=1}” and
“corresponding to the case {k=2}” should read ” … >= 0 ” ?

WFL

[Corrected, thanks – T.]



Suvrit Sra (@optiML) · 1 October 2017

A minor clarification regarding page 35 of your paper. It says that “the conjecture – as stated in [8] and proved in [34] – has a minor error, namely when $\sum_j m_j > \sum_j n_j$” – In [34] the conjecture is proved for true majorization, where $\sum_j m_j = \sum_j n_j$ holds. Or am I missing something? Thanks!



Terence Tao · 2 October 2017

Thanks for this, it looks like we did not quite quote the CGS conjecture correctly, and will sort this out in the next revision of the ms.



Rachid Ait-Haddou · 1 October 2017

Dear Terence,
Will it be OK to change my first name to Rachid instead of Rachit :). Many thanks for the mention. Also as Sra just mentioned, what is proved in the paper is slightly different from Cuttler-Greene-Skandera conjecture. The paper proves that if $m_i \geq n_i$ for all i then

the inequality between the normalized Schur functions holds. The conjecture states that if $\sum_{k=1}^{j} m_k \geq \sum_{k=1}^{j} n_k, j = 1, 2, ..., N-2$ and $\sum_{k=1}^{N-1} m_k = \sum_{k=1}^{N-1} n_k$ then the inequality between the normalized Schur functions holds.



Terence Tao · 2 October 2017

Sorry for the typo! We'll correct the attribution to the CGS paper in the next revision of the ms.



Rachid Ait-Haddou · 1 October 2017

It is sometimes confusing to go from powers to partitions (when it is done hazily). I would like to correct the statement of the conjecture in terms of the powers. It states that if $\sum_{k=j}^{N-1} m_k \geq \sum_{k=j}^{N-1} n_k, j = 1, 2, ..., N-1$ and $\sum_{k=0}^{N-1} m_k = \sum_{k=0}^{N-1} n_k$ then the inequality between the associated normalized Schur functions holds over $]0, \infty[^N$. I hope I got it right this time.



Aula · 2 October 2017

How does Theorem 1 say anything about the value of the expression ah-dg? If I understood the definitions correctly, it should probably be ah-de instead.

[Corrected, thanks -T.]



A. Barreras · 16 October 2017

Dear Terence,
In Linear Algebra literature, there exist two "school of notations":
1: TP: matrices with non-negative minors
STP: matrices with strictly positive minors

2: TN: matrices with non-negative minors
TP: matrices with posive minors

If I understood correctly, you are using the first one. In this case, in the definition of totally positive (p. 38 of the article, line -3), should be changed formula "det(A_{I,J})>0" by "det(A_{I,J}) \geq 0"?

Sorry if this is just my misunderstood.

1. Allan Pinkus (2009), Totally Positive Matrices, Cambridge University Press, ISBN 9780521194082
2. Fallat, S. M., & Johnson, C. R. (2011). Totally nonnegative matrices. Princeton University Press.



Terence Tao · 17 October 2017

In our paper we're using total positivity in the strict sense. (A large part of the paper is concerned instead with positive semi-definite matrices, though.)